

Advanced Statistics for Social Sciences 1

Statistical Thinking and Data Analysis with R

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Logistics

Possible re-schedule of next week's class

Fill out SET Bruin survey

Outline

Last class

- Compute and interpret effect sizes
- Understand confidence intervals
- Bayes factor
- Modeling categorical relationships

Today

- Modeling continuous relationships
- Covariance and correlation
- Linear regression
- Best practices and replicable science

What can statistics do for us?

DESCRIBE

- Summarize data
- Visualize patterns
- Estimate parameters

DECIDE

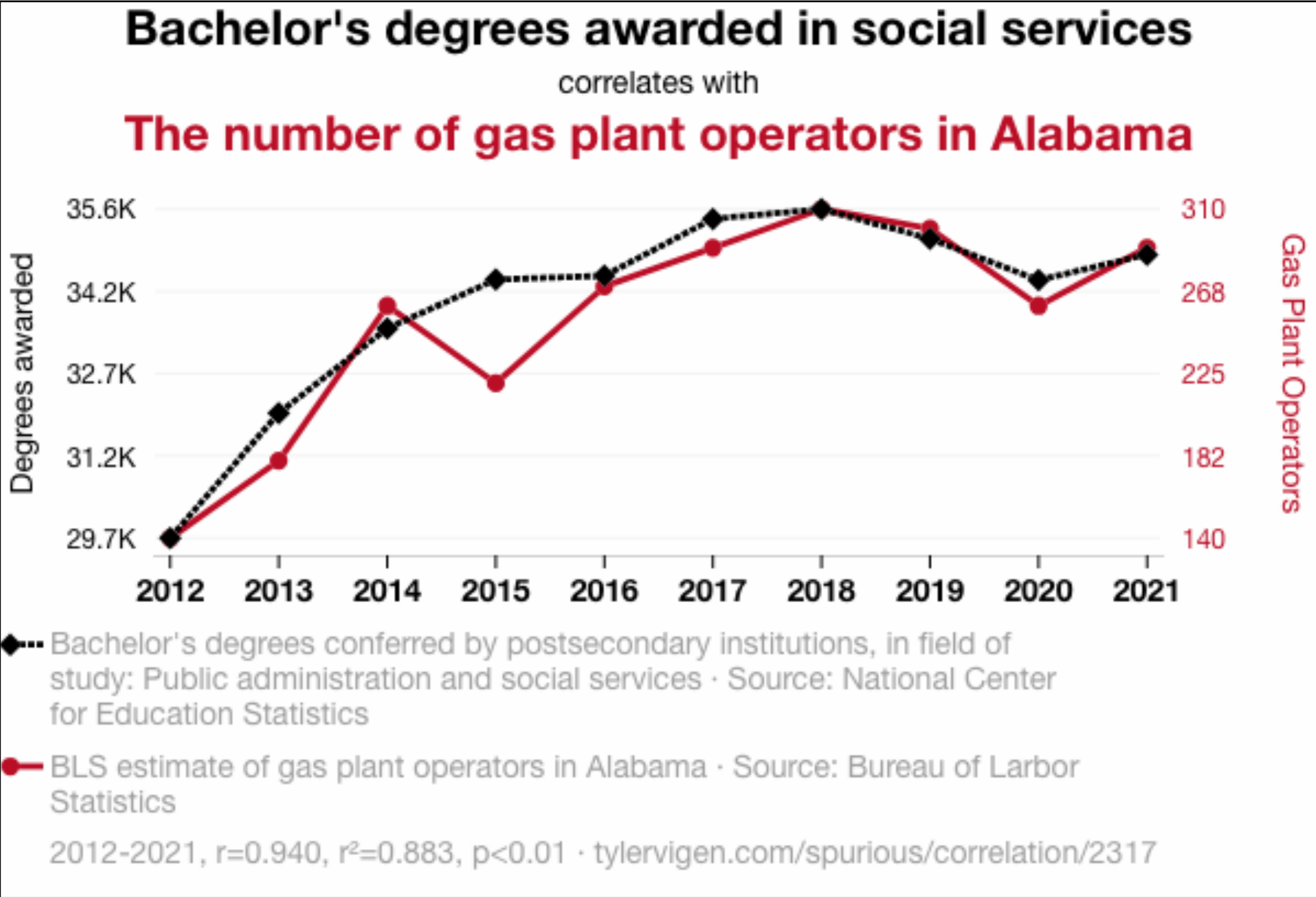
- Test hypotheses
- Compare groups
- Evaluate claims

PREDICT

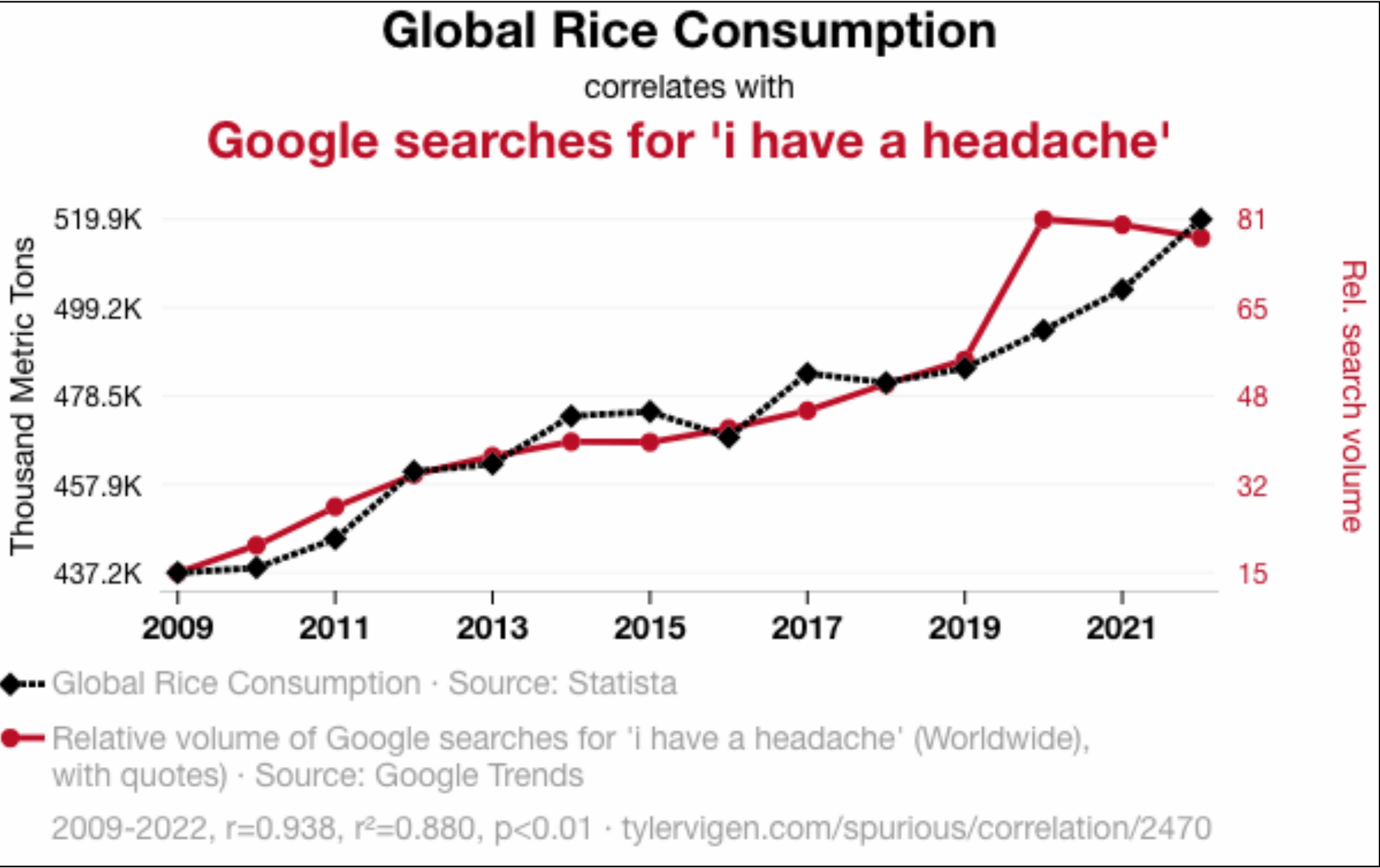
- Build models
- Forecast outcomes
- Classify new data

Correlations

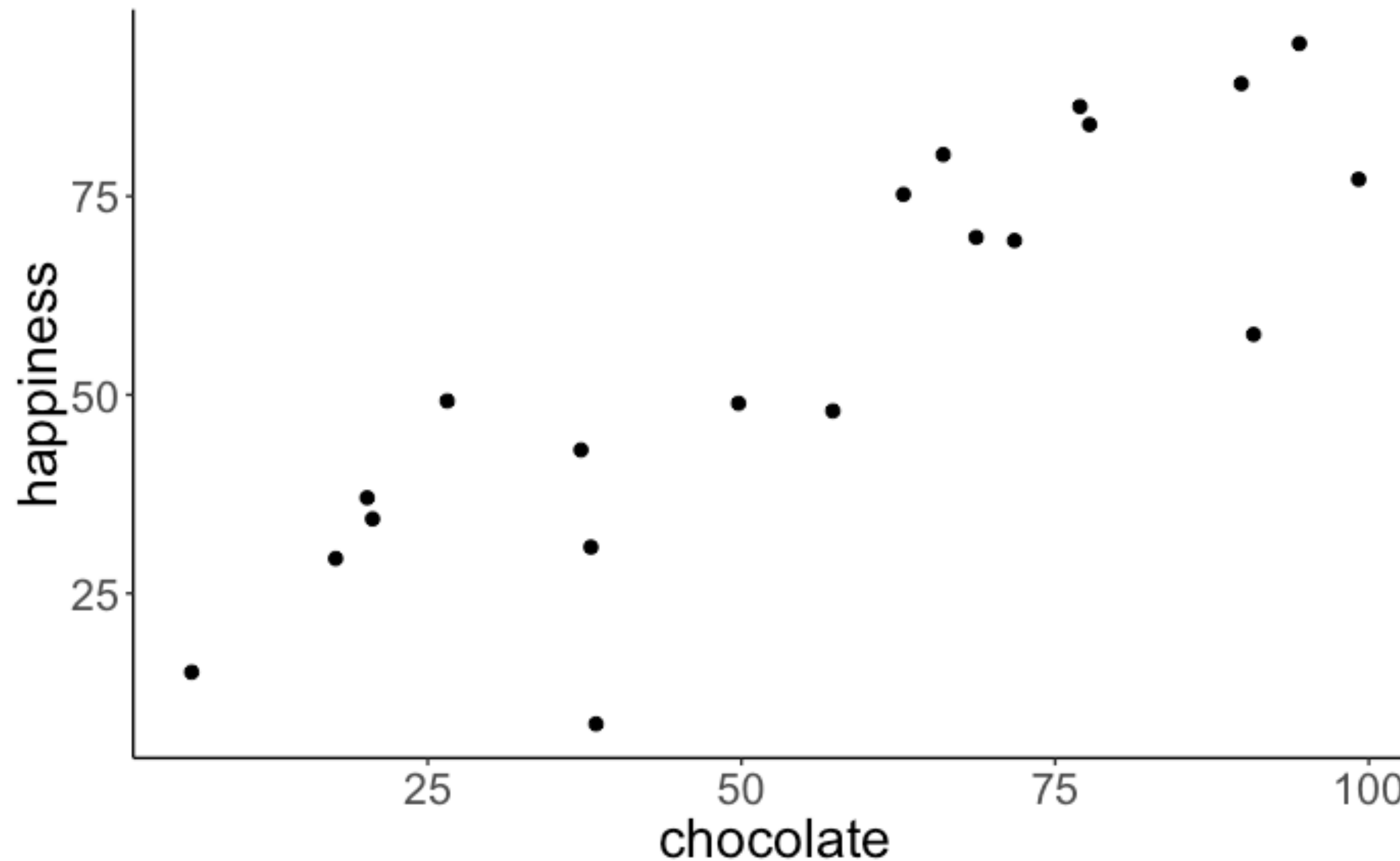
Gas plants in Alabama are making everyone study social science..
Valid conclusion?



We should stop eating rice..
Valid conclusion?



How to best characterize the relationship
between x and y by a single number?



correlation = a measure of the relationship
between two variables

Covariance

Recap: Variance captures how far individual data points are from the mean, on average

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{N - 1}$$

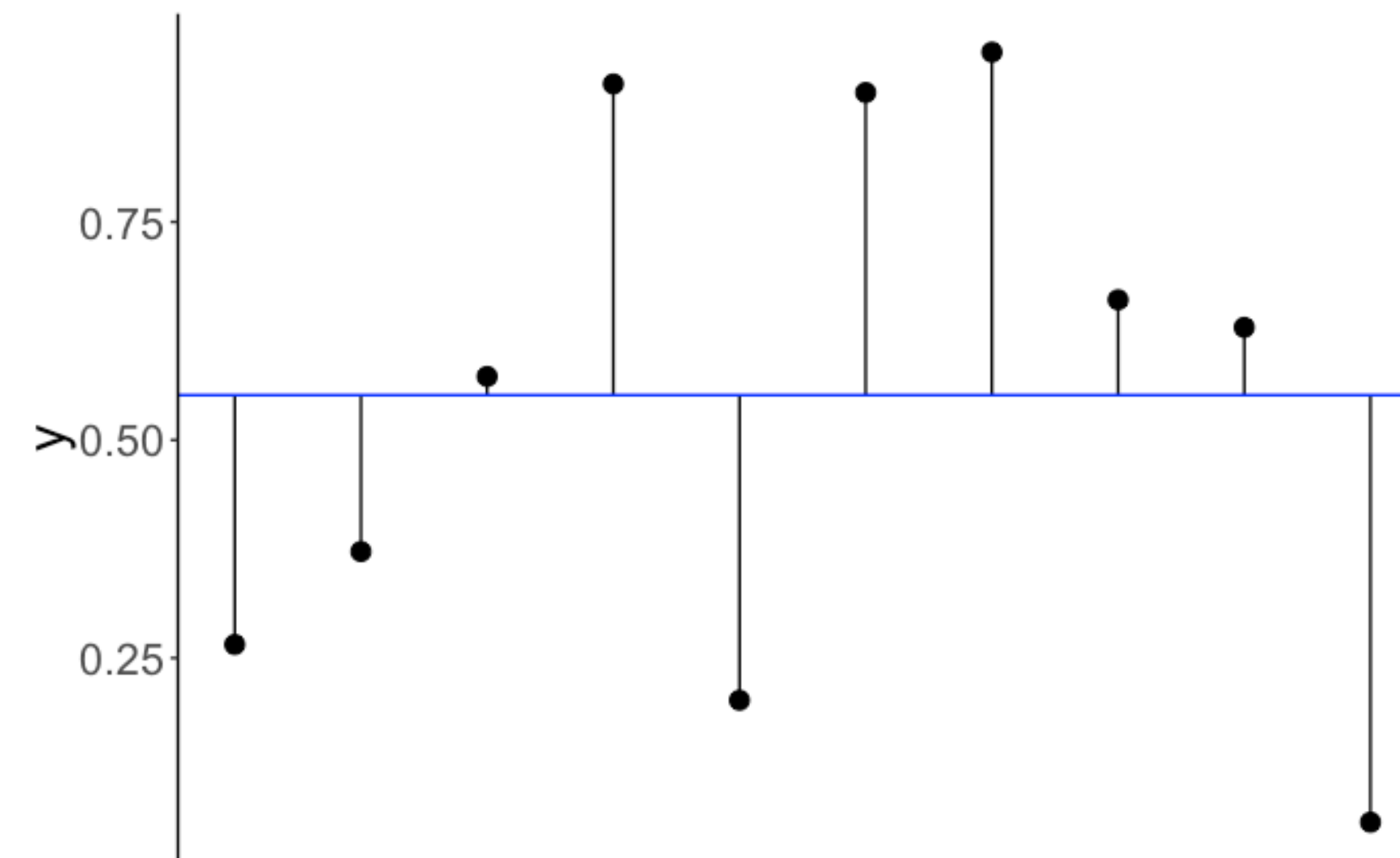
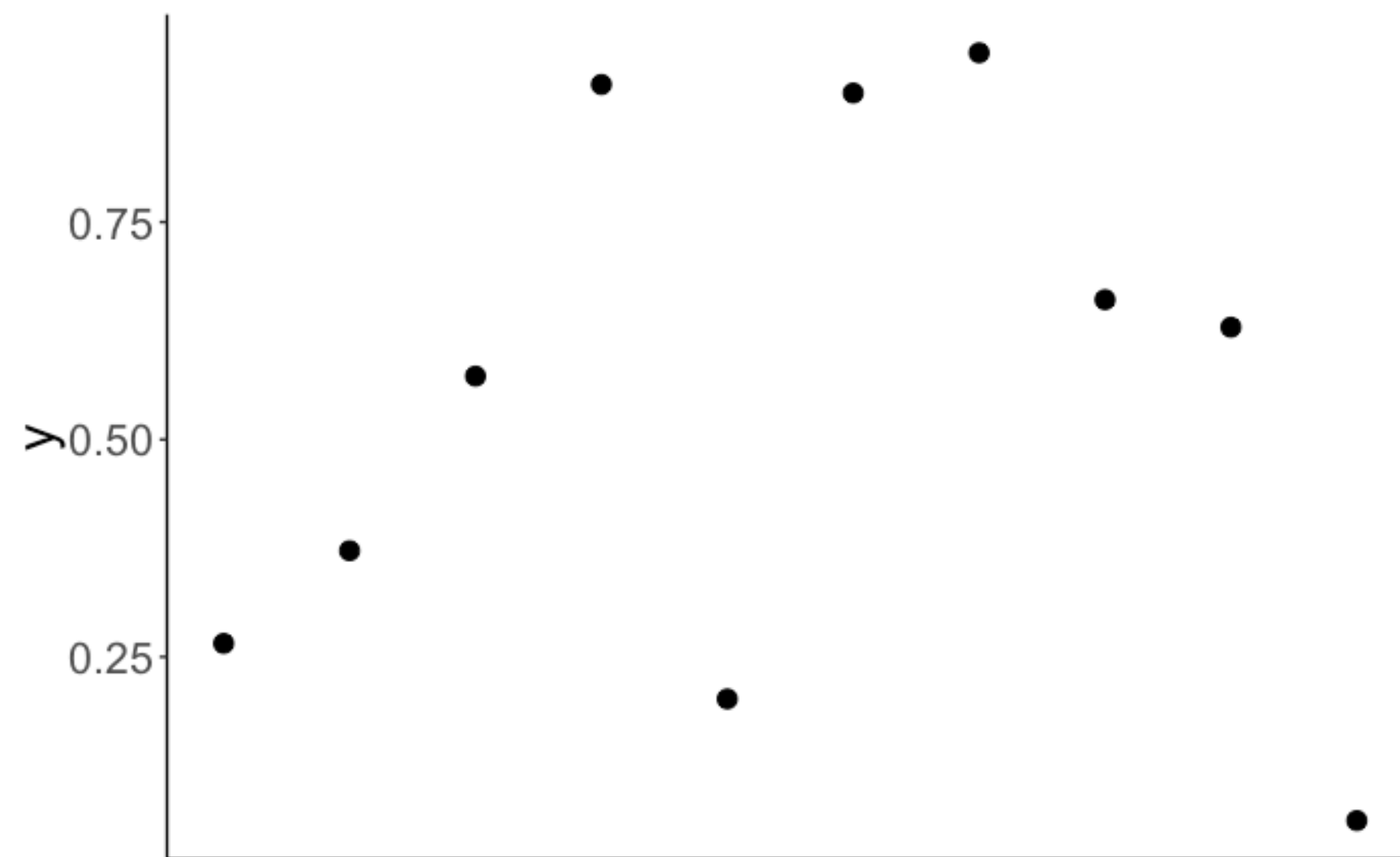
Covariance: Is there a relationship between the deviations of two different variables across observations?

$$covariance = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

sample variance

$$Var(X) = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n - 1}$$

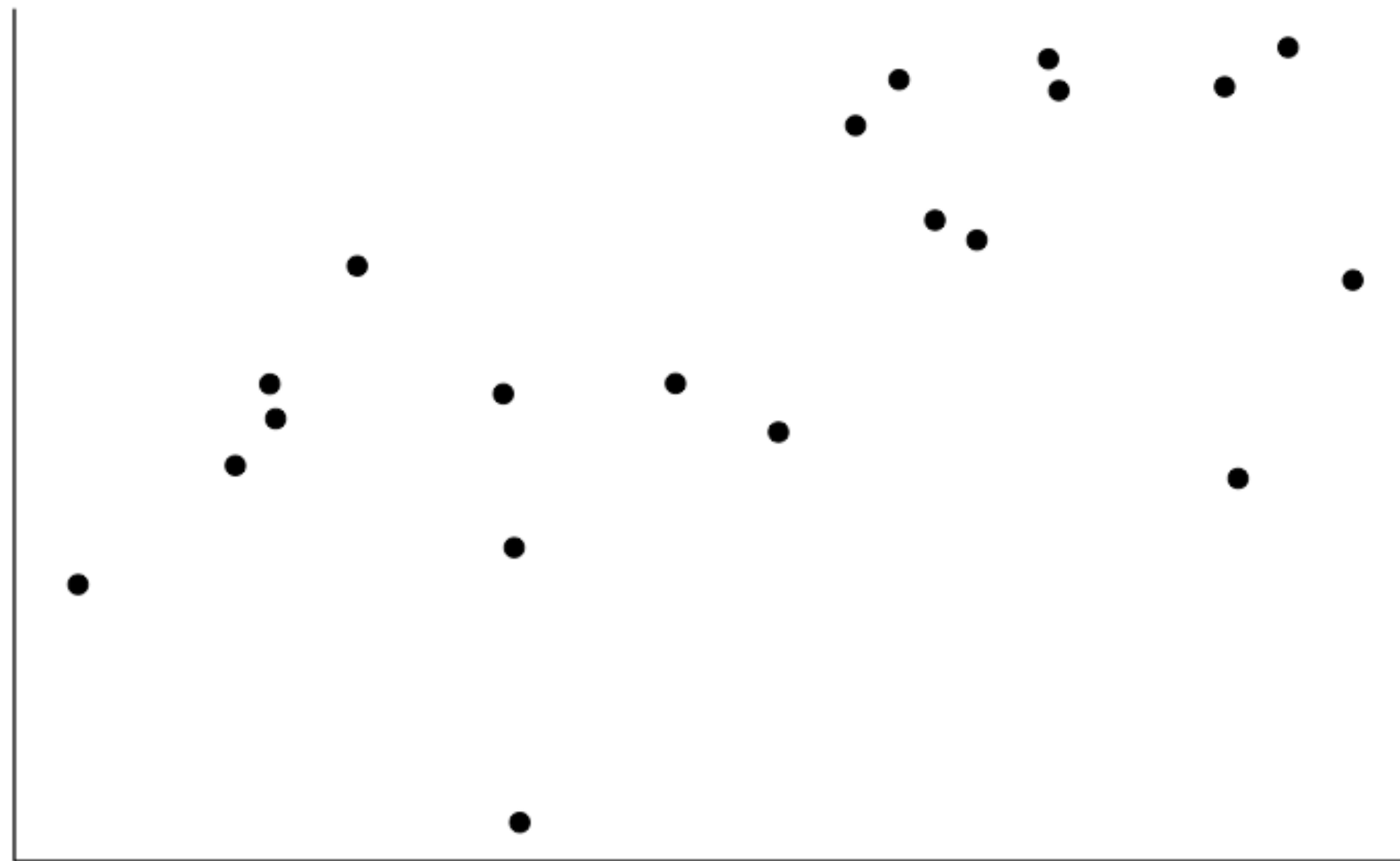
sum of squared errors



How well does the mean capture the data?

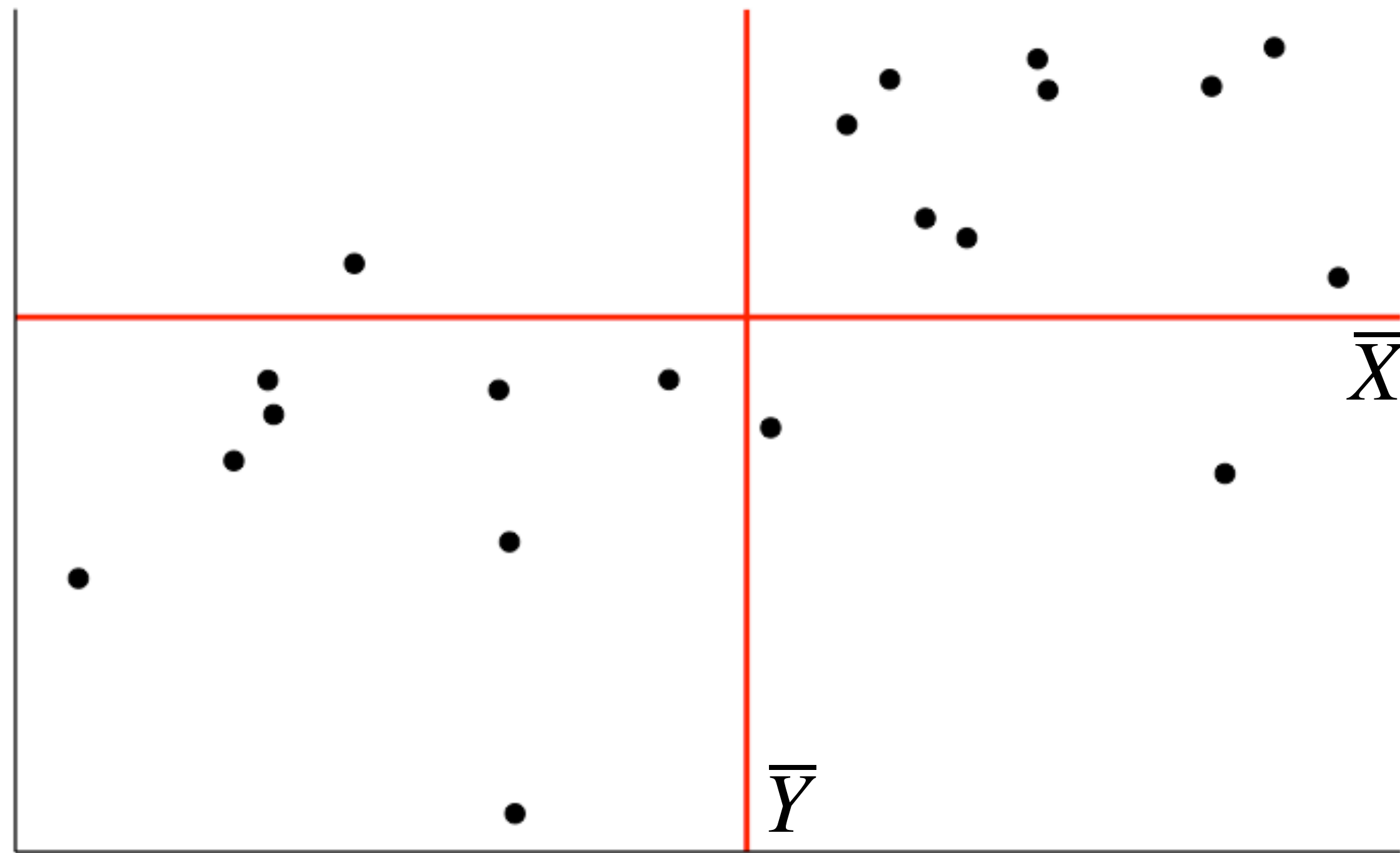
sample covariance

$$\text{Cov}(X, Y) = \sum_{i=1}^n \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$



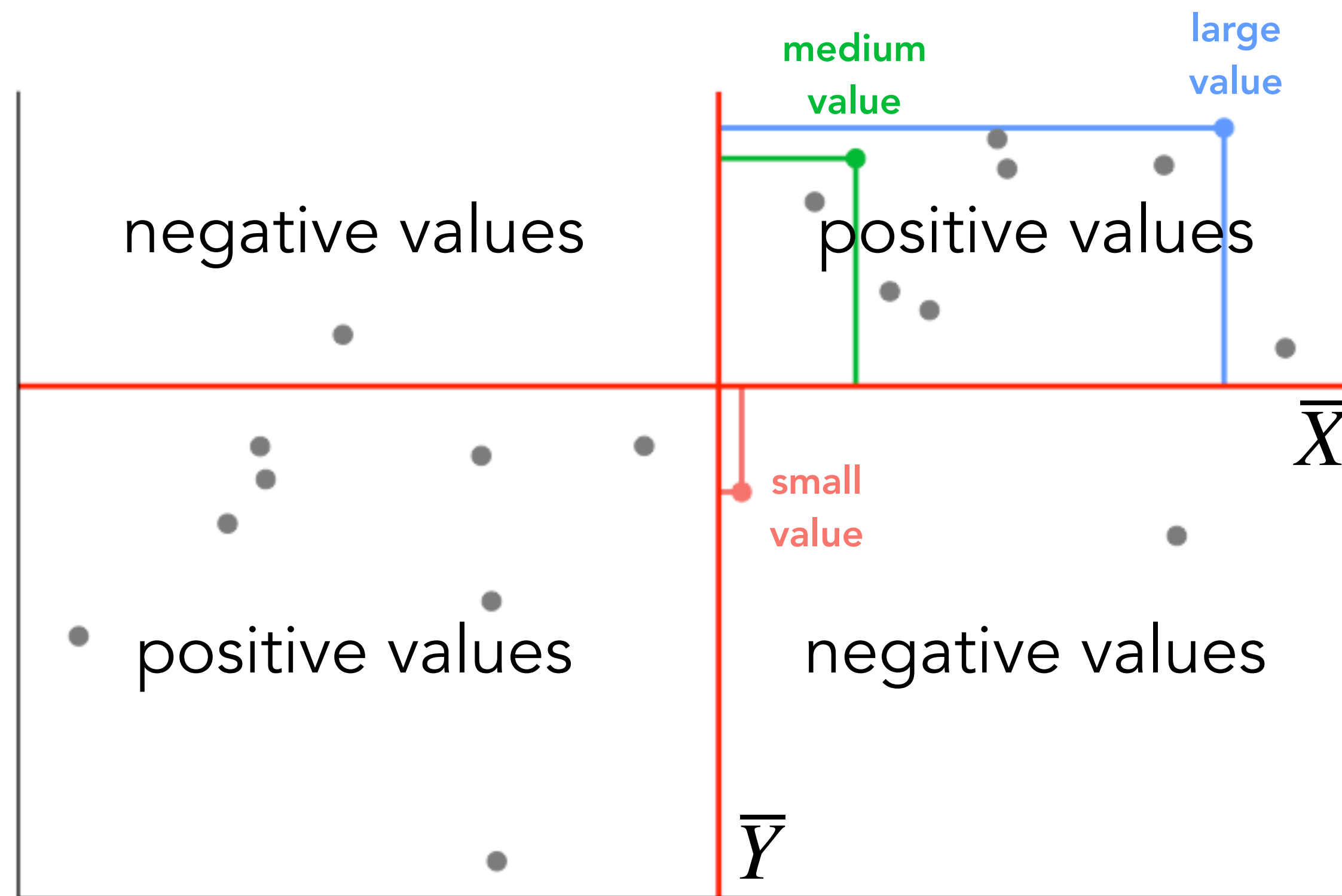
sample covariance

$$\text{Cov}(X, Y) = \sum_{i=1}^n \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$



sample covariance

$$Cov(X, Y) = \sum_{i=1}^n \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$



sample covariance

$$Cov(X, Y) = \sum_{i=1}^n \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- Value is far from zero when individual data points deviate by similar amounts from their means
- If they are deviant in the same direction then the covariance is positive
- If they are deviant in opposite directions the covariance is negative

sample covariance

$$\text{Cov}(X, Y) = \sum_{i=1}^n \frac{(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

depends on the scale of the variables.. we can easily fix this!

the $n - 1$ s cancel out

sample correlation coefficient

$$r(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \cdot \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$


standardized covariation
(dividing by the standard deviations)

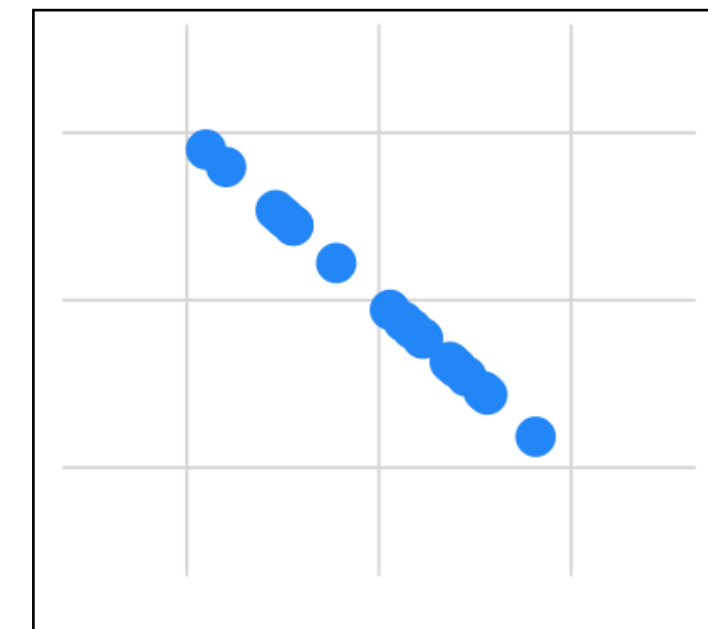
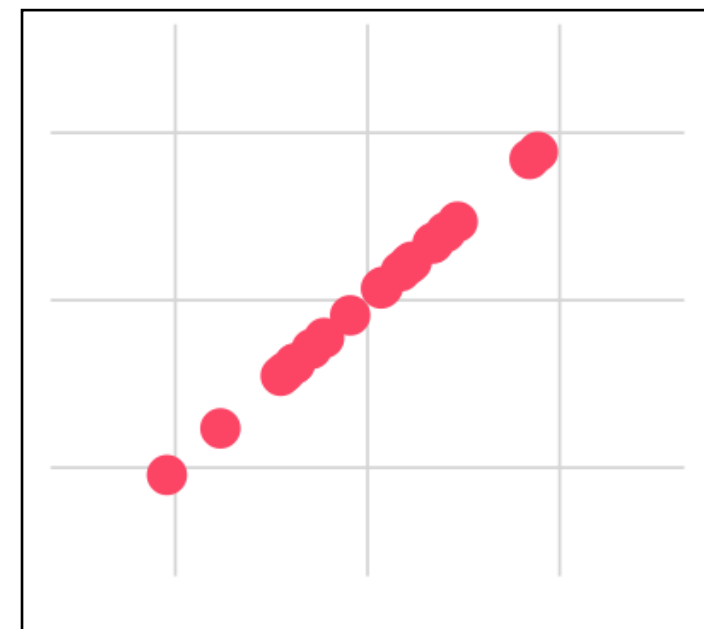
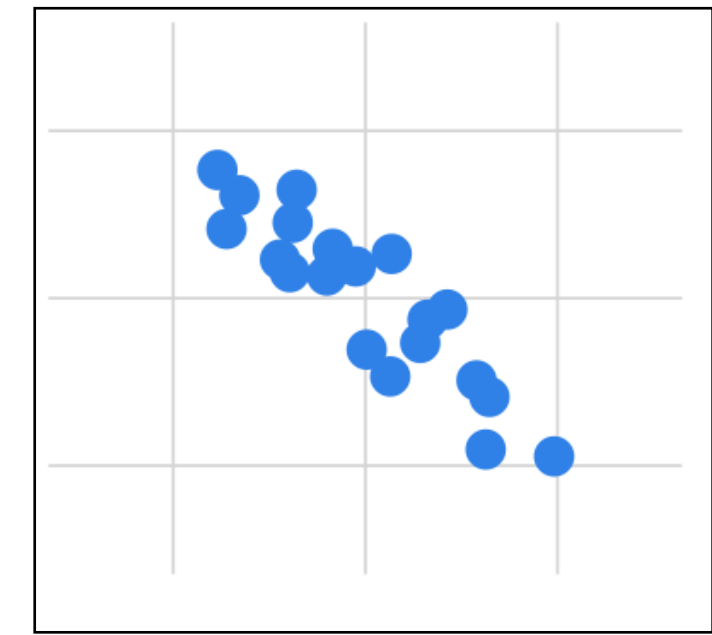
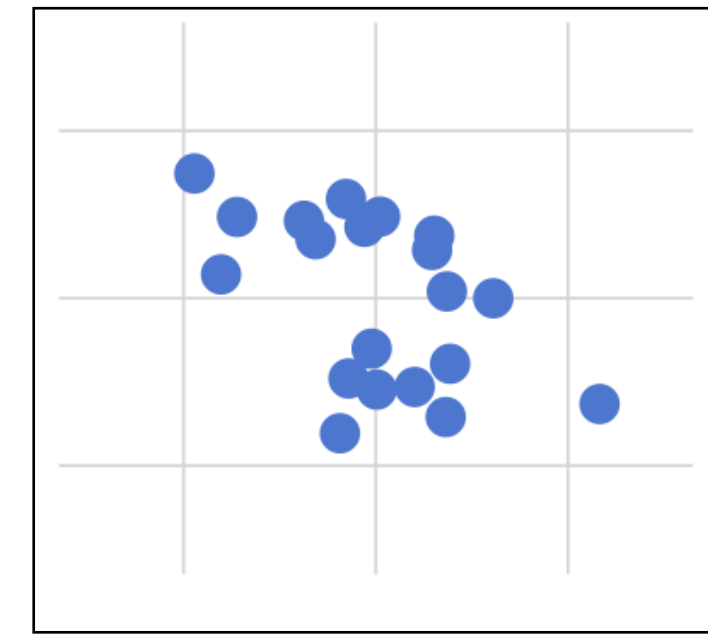
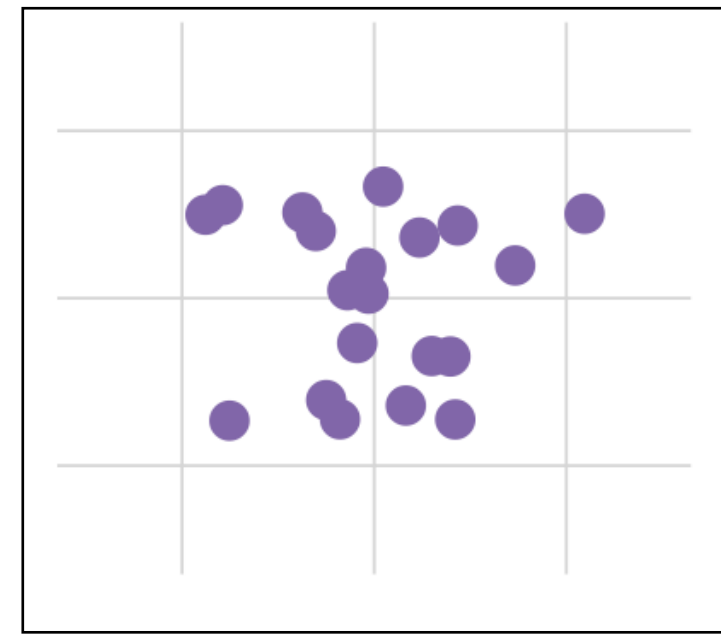
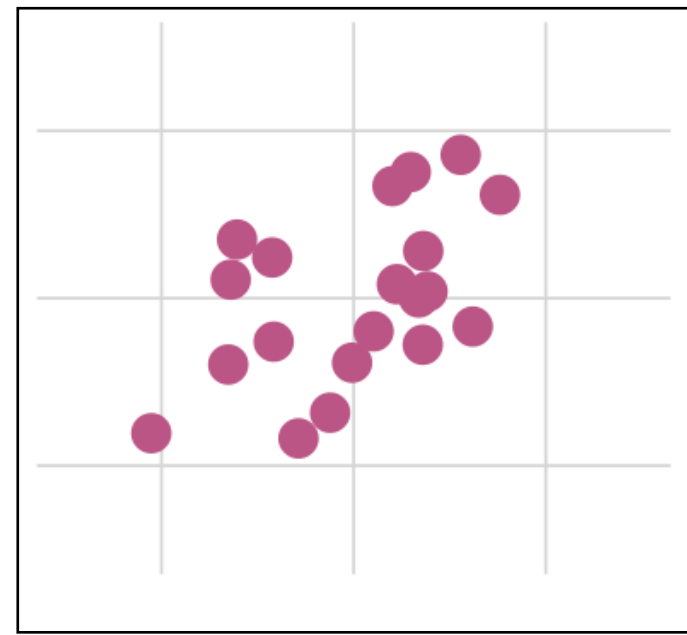
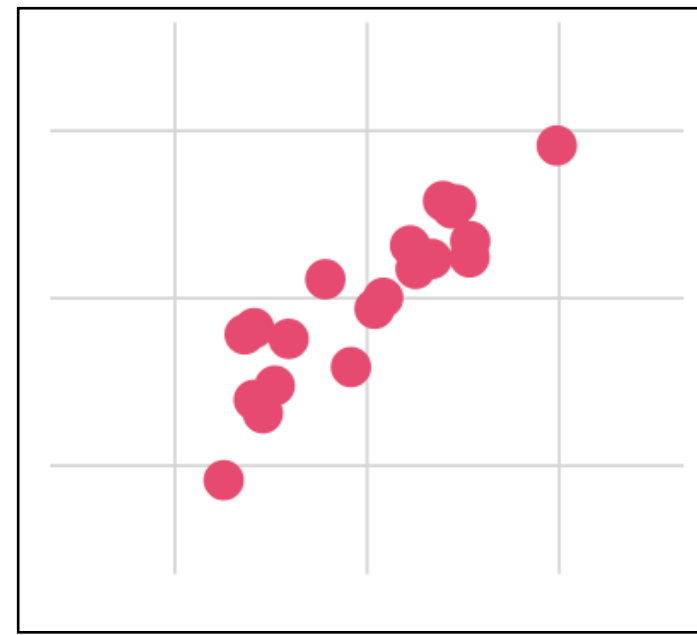
How to best characterize the relationship between x and y by a single number?

Pearson's correlation

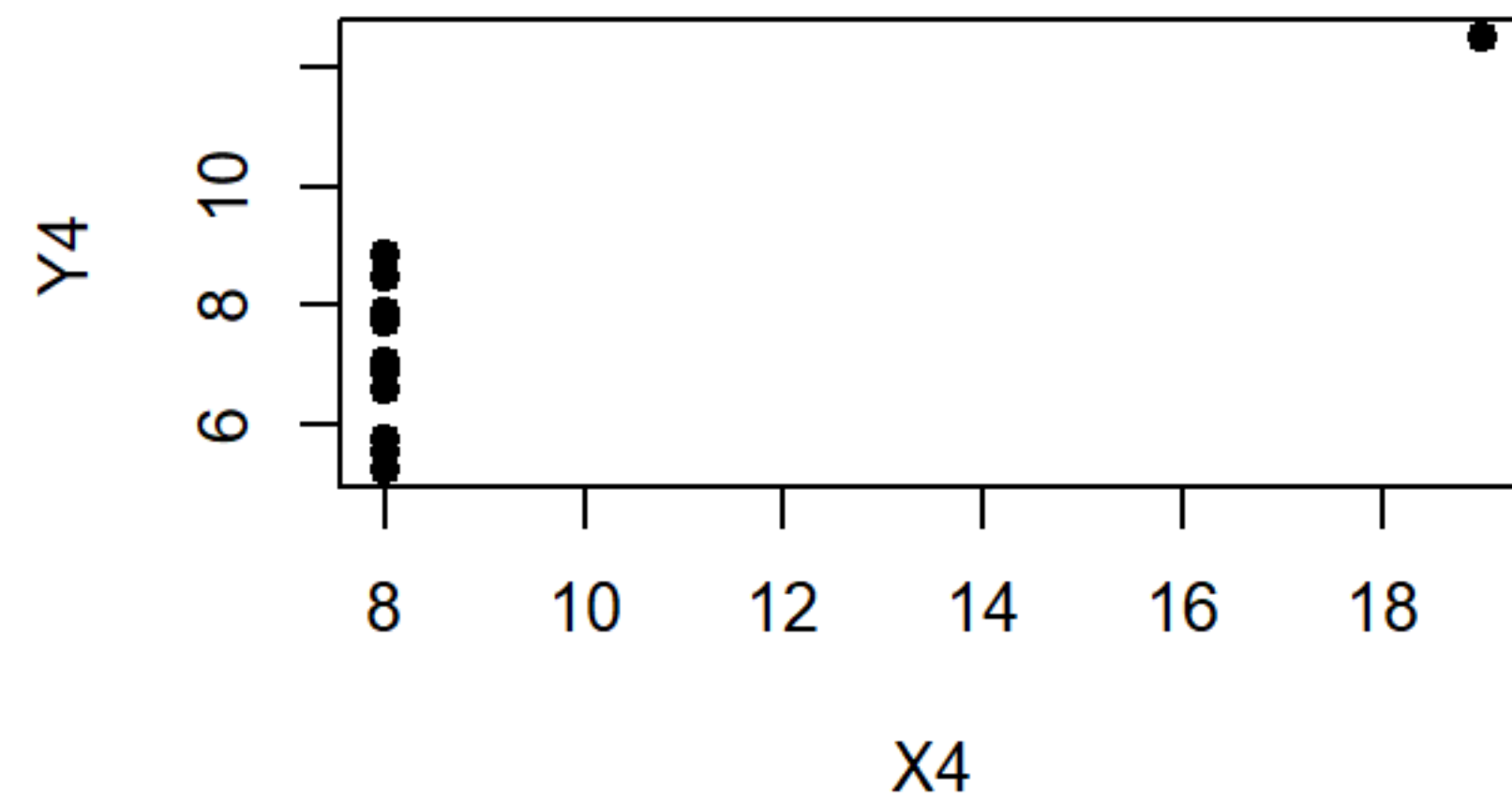
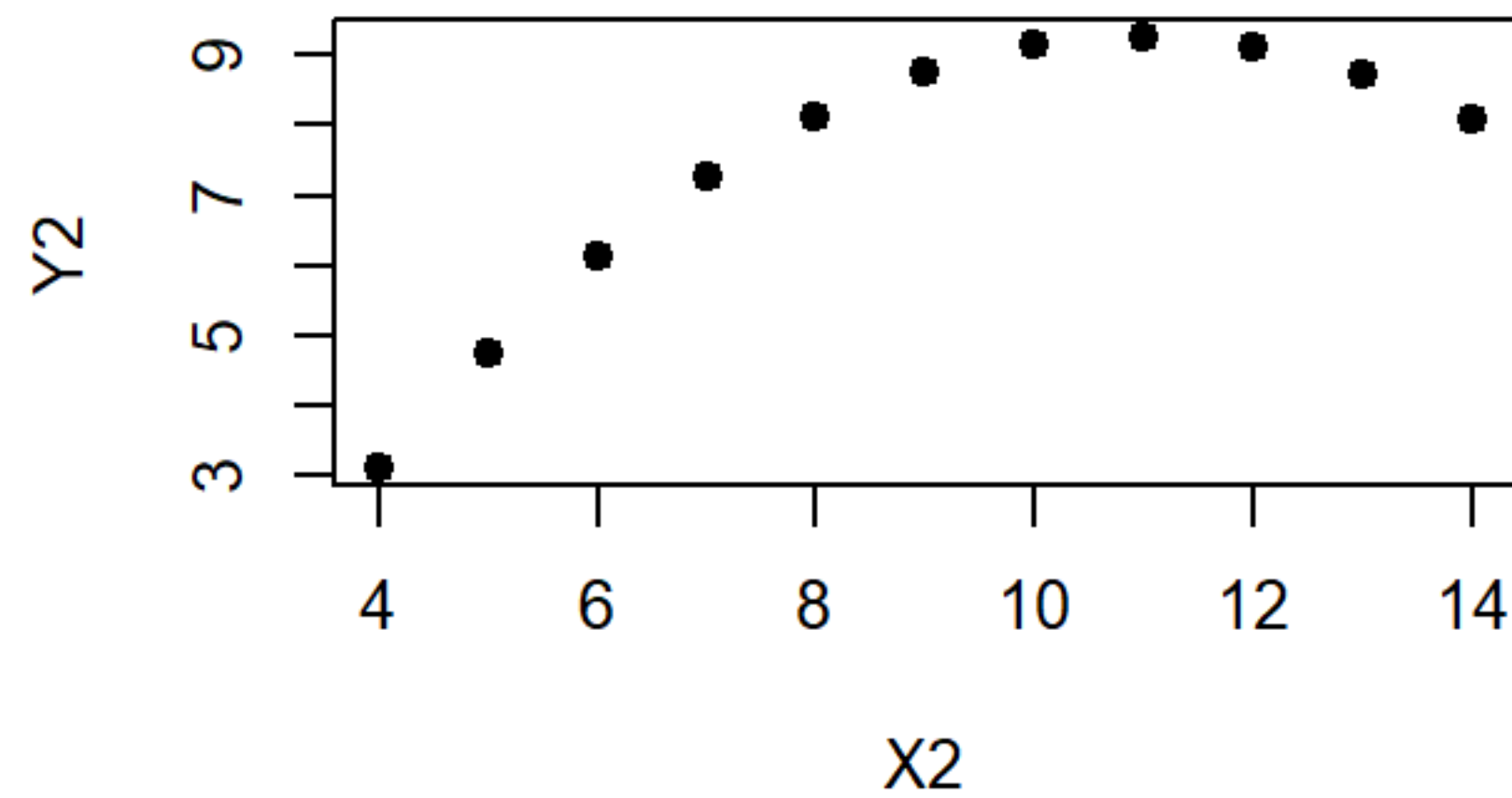
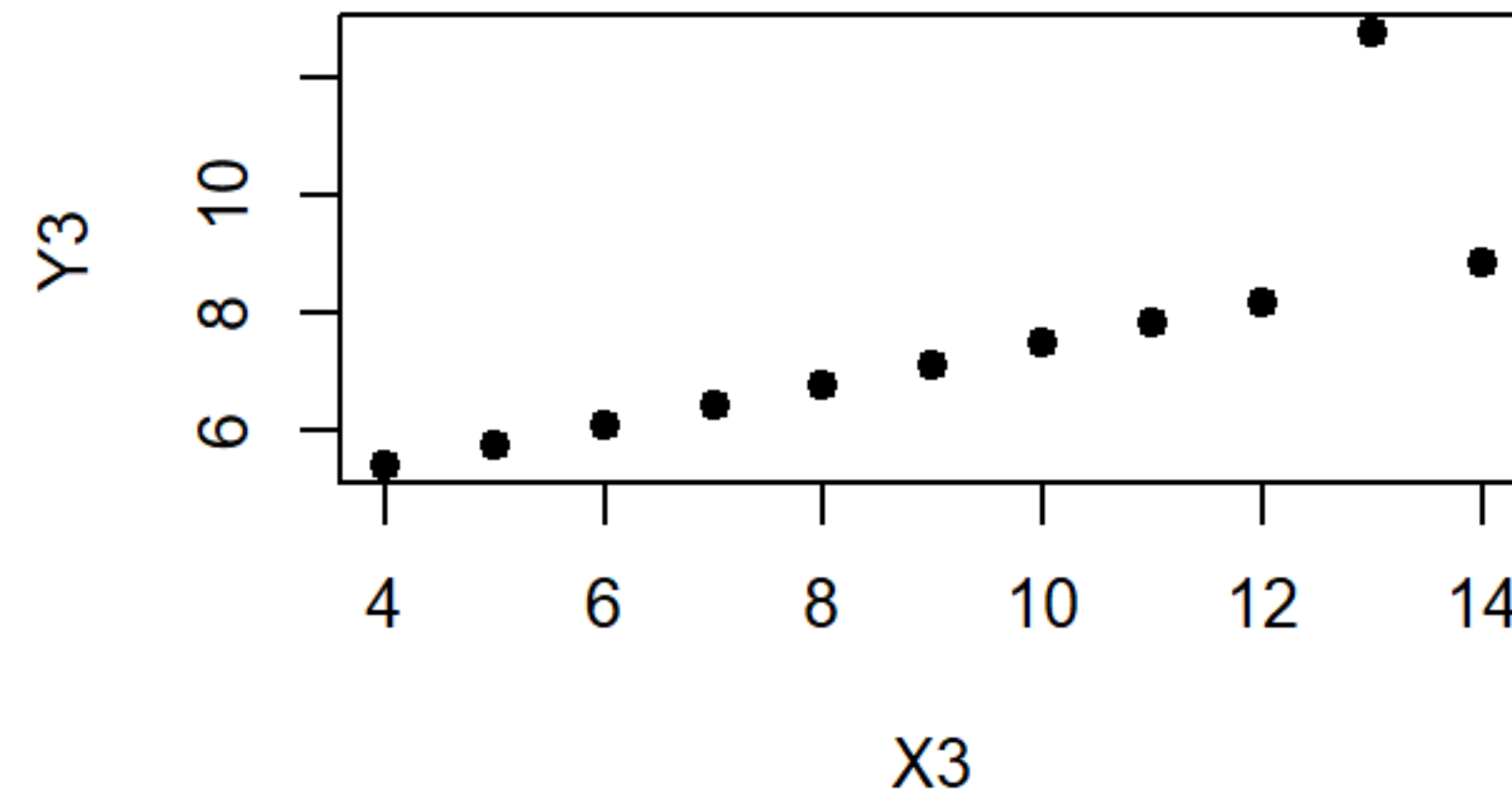
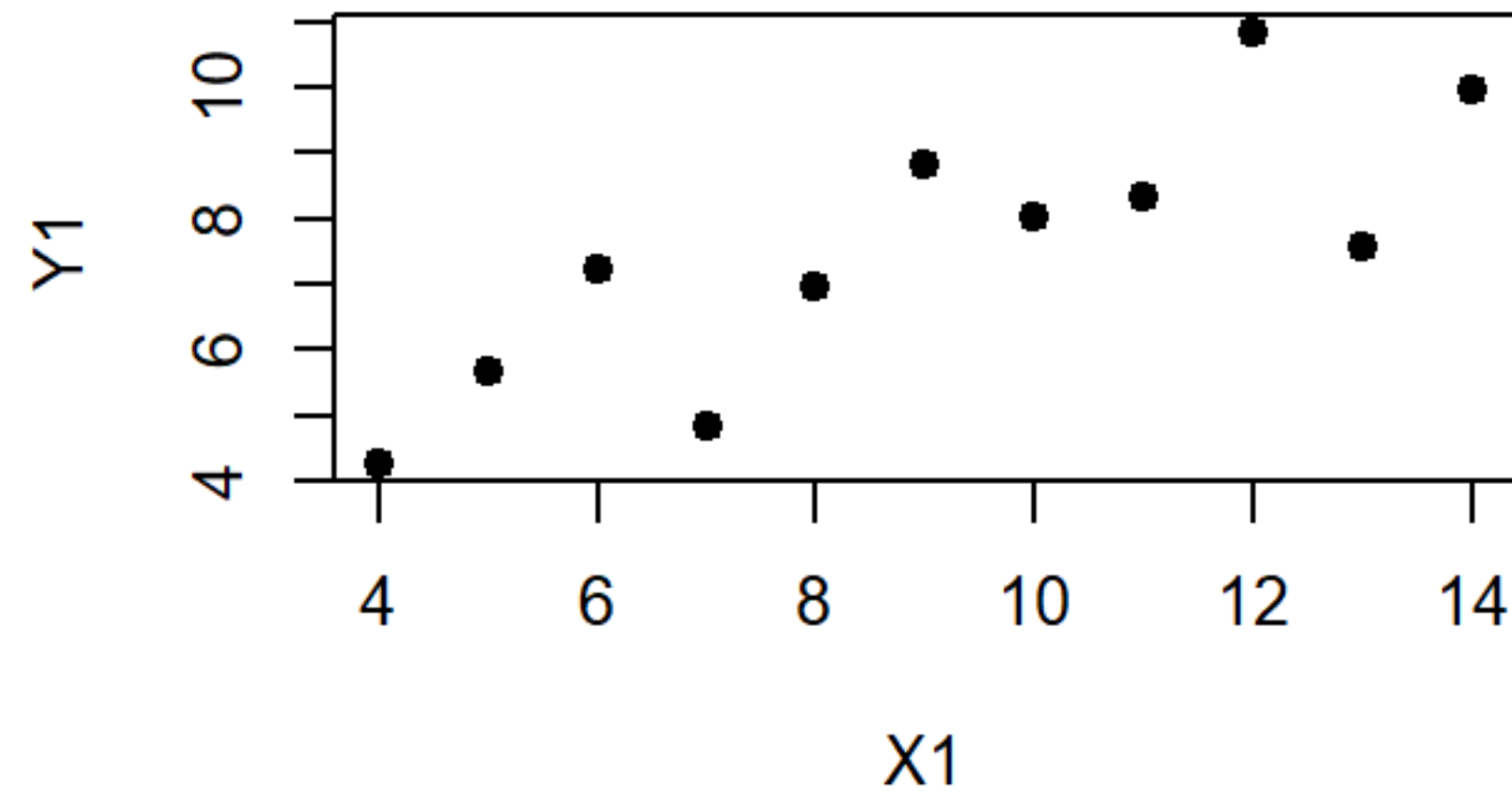
$$r = \frac{\textit{covariance}}{S_x S_y}$$

- standardized: $-1 \leq r \leq 1$
- scale independent (for both X and Y)
- commutativity: $r(X, Y) = r(Y, X)$
- sign determines the direction of dependence
- captures **linear dependence** only

How to best characterize the relationship between x and y by a single number?



Always graph your data!

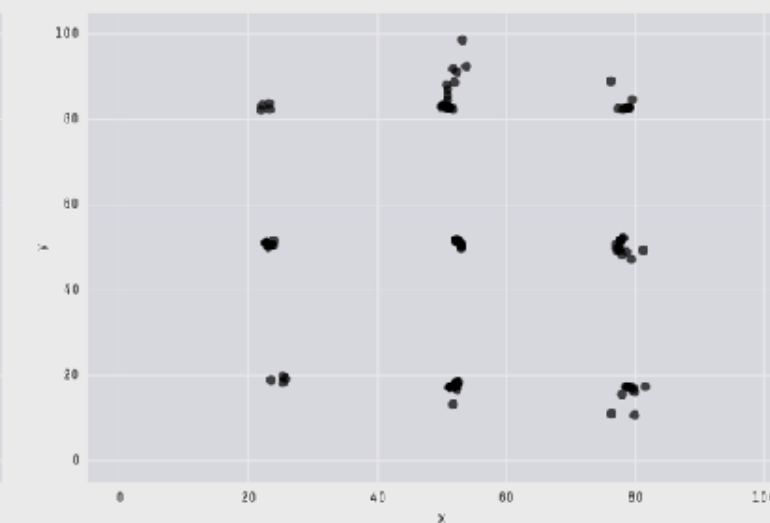
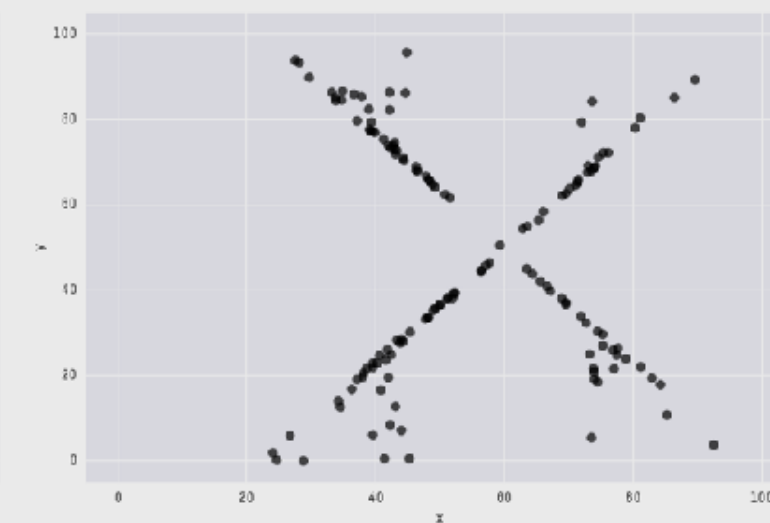
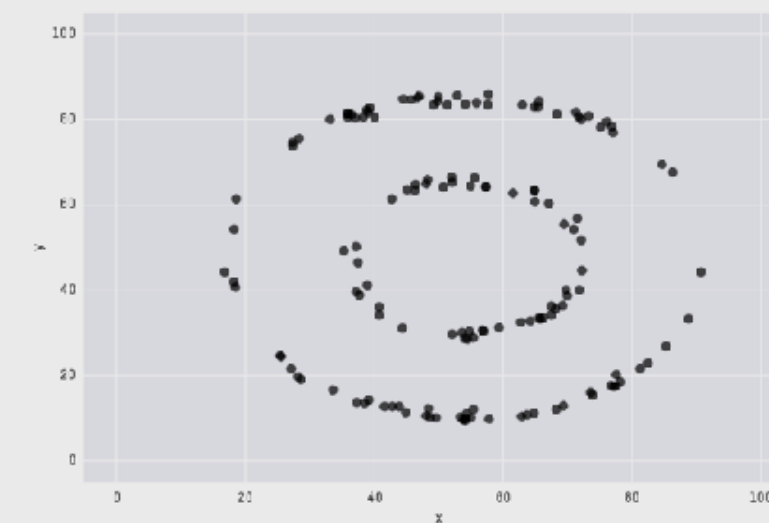
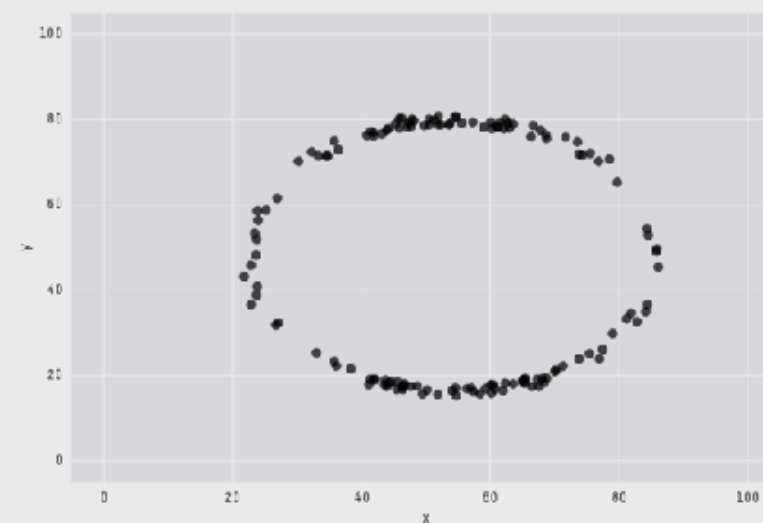
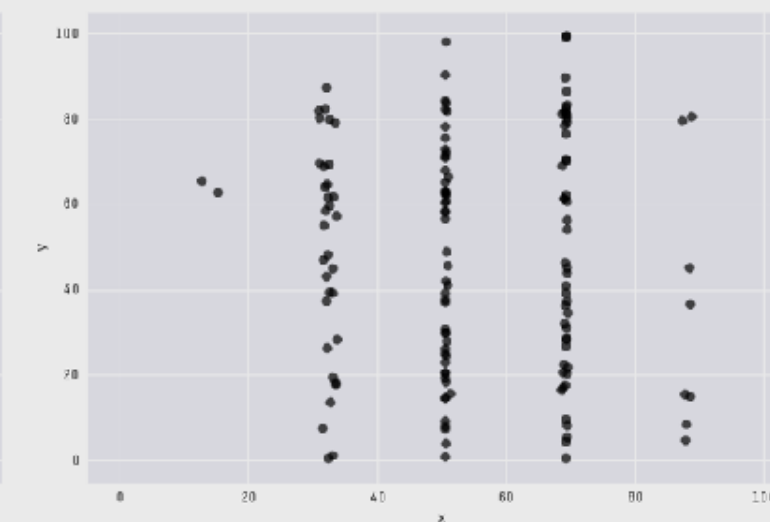
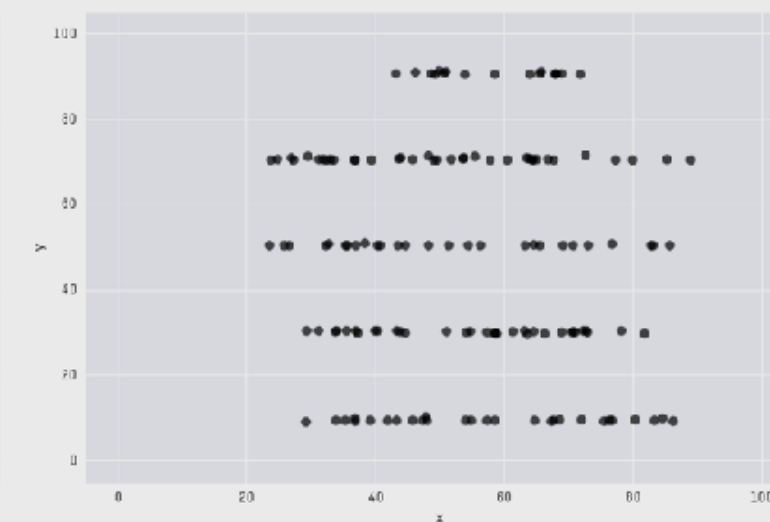
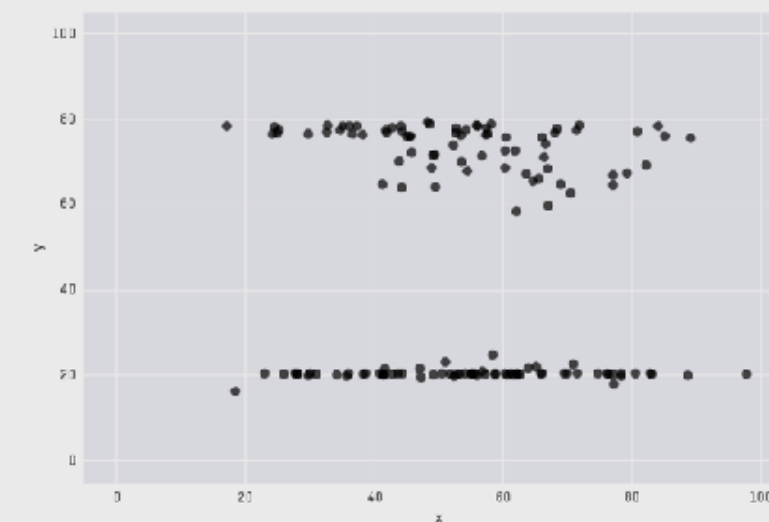
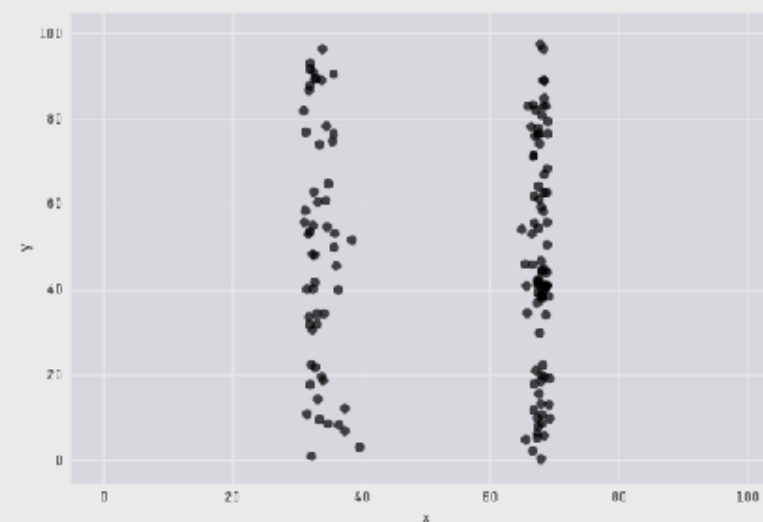
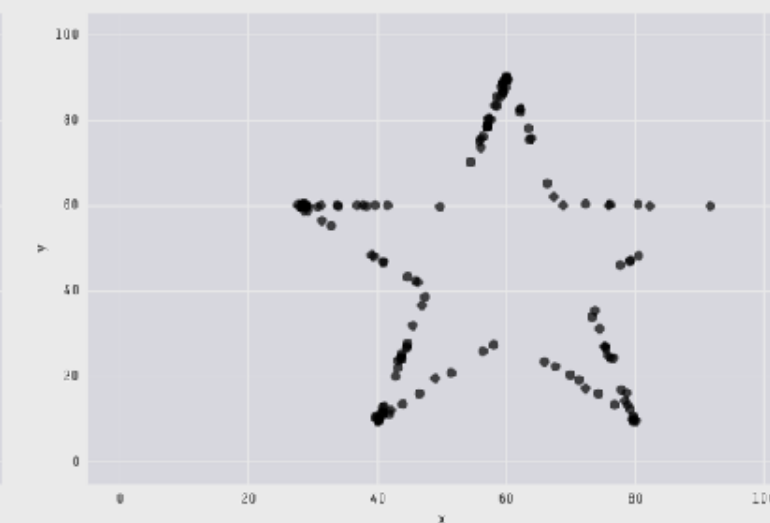
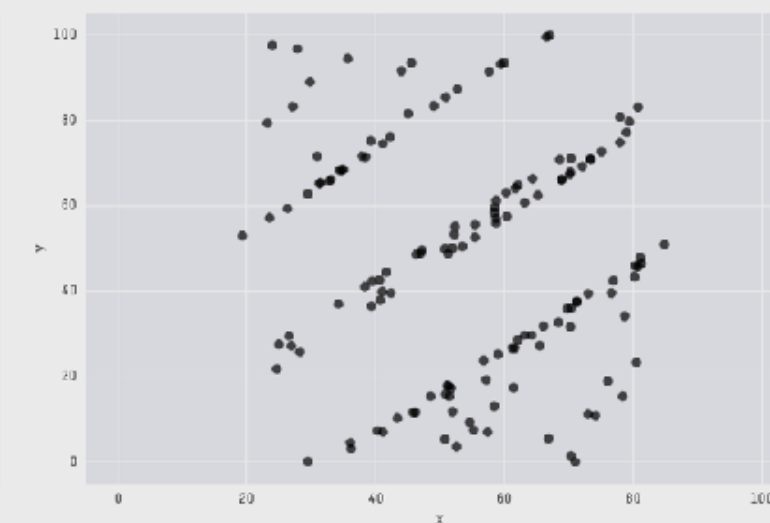
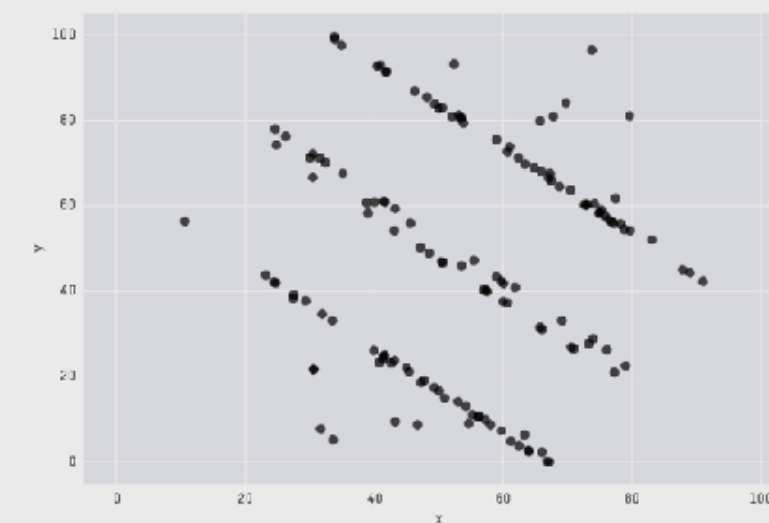
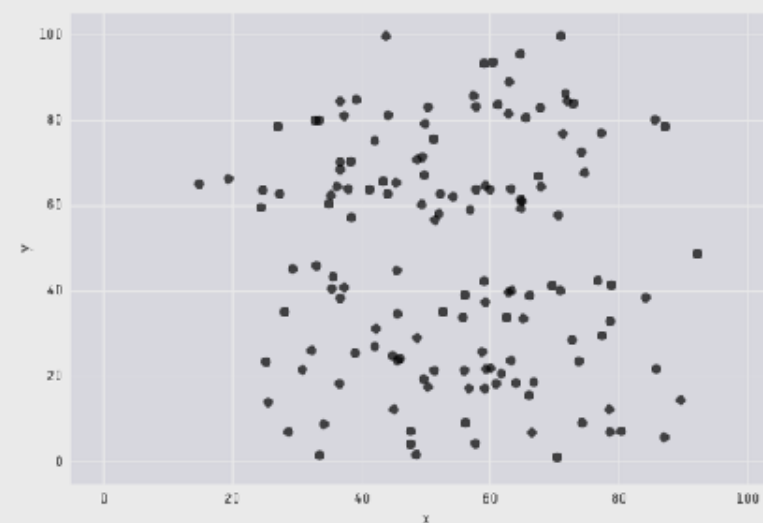


Anscombe's quartet. All four of these data sets have a Pearson correlation of $r = 0.816$!

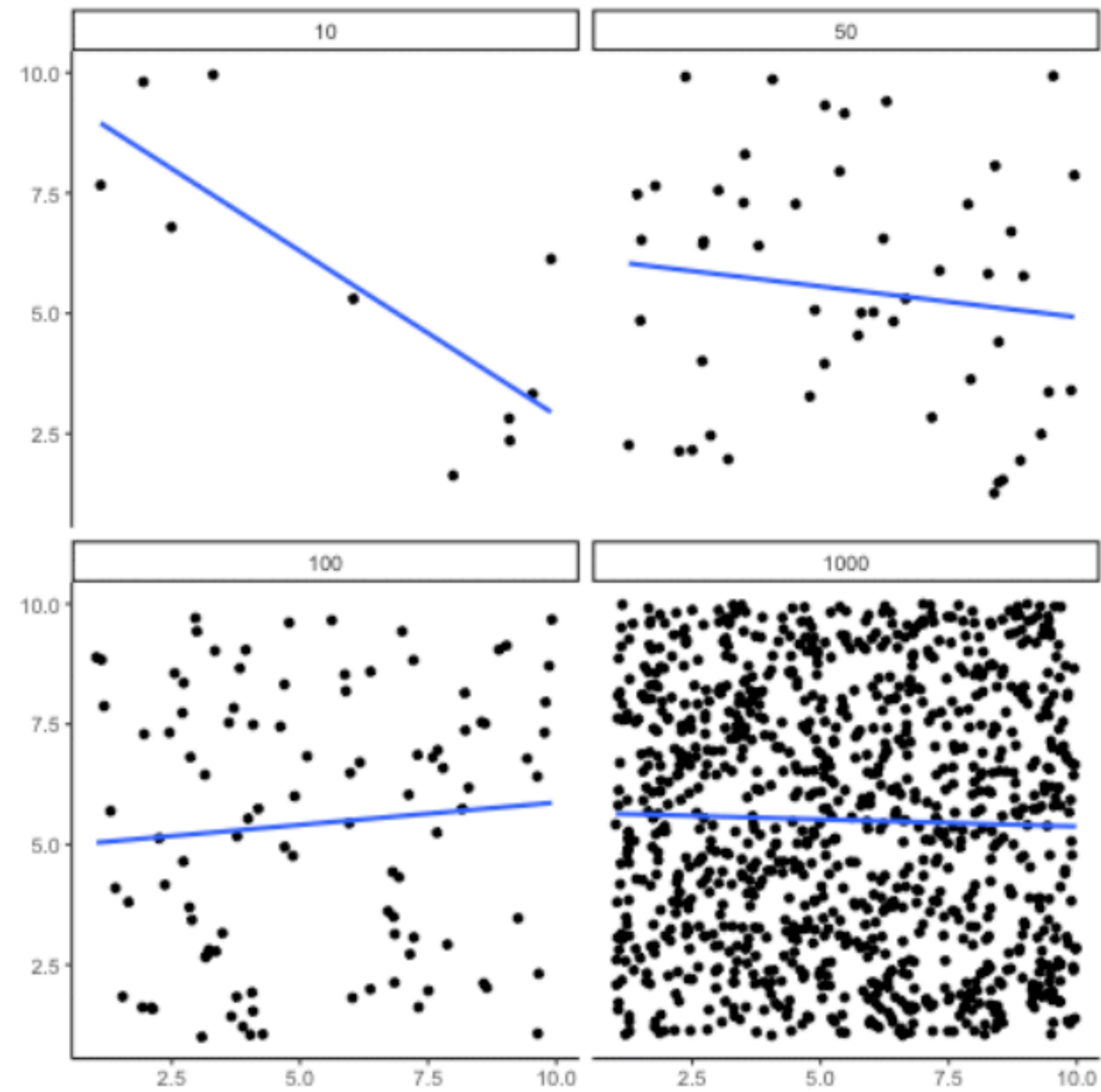
Always graph your data!



X Mean: 54.26
Y Mean: 47.83
X SD : 16.76
Y SD : 26.93
Corr. : -0.06



Beware of small samples

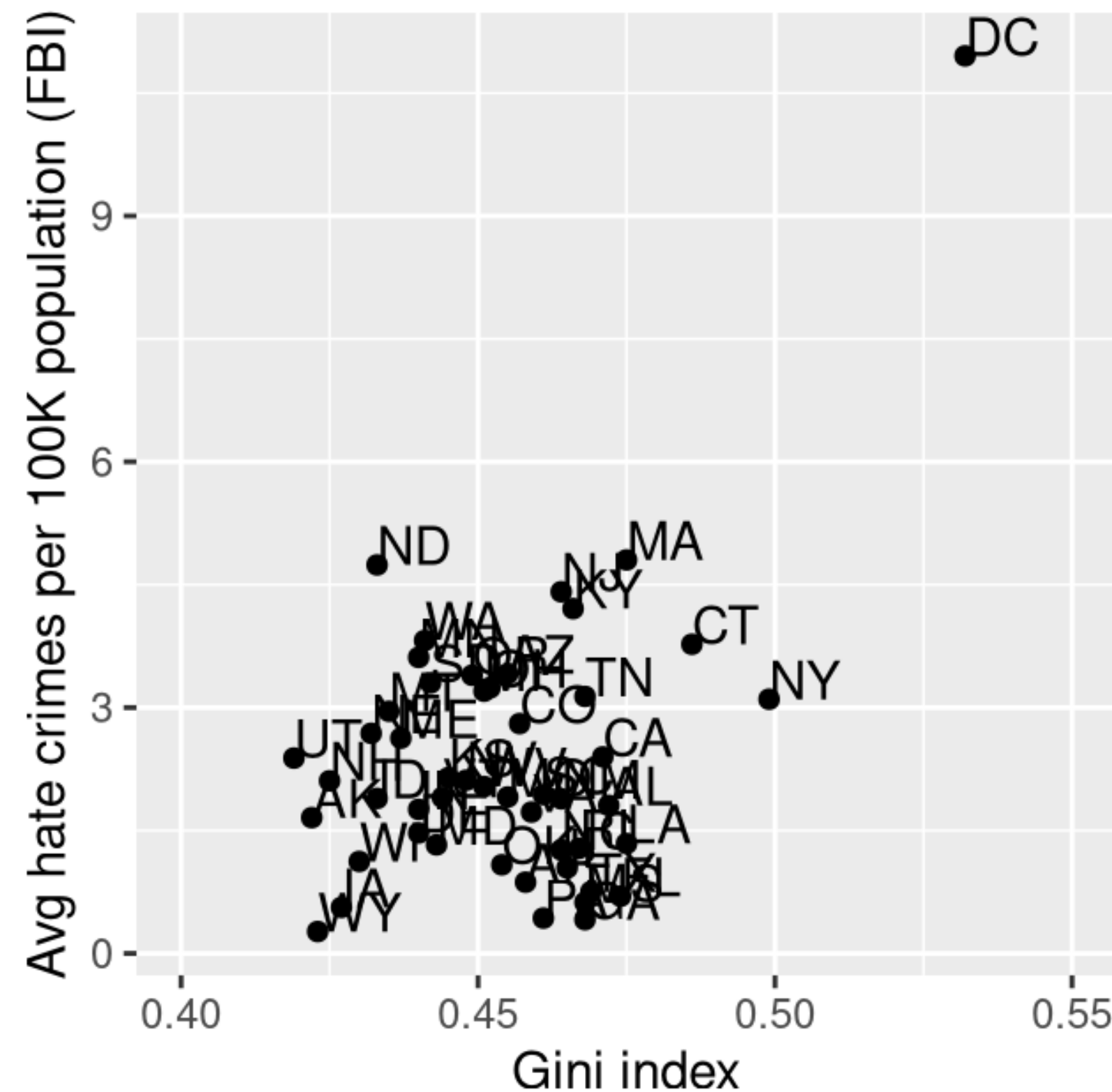


Beware of outliers!

Let's use an example to understand this..

Higher Rates Of Hate Crimes Are Tied To Income Inequality (Fivethirtyeight.com, 2017)

“We found that income inequality was the most significant determinant of population-adjusted hate crimes and hate incidents across the United States”.



Is income inequality related to hate crimes?

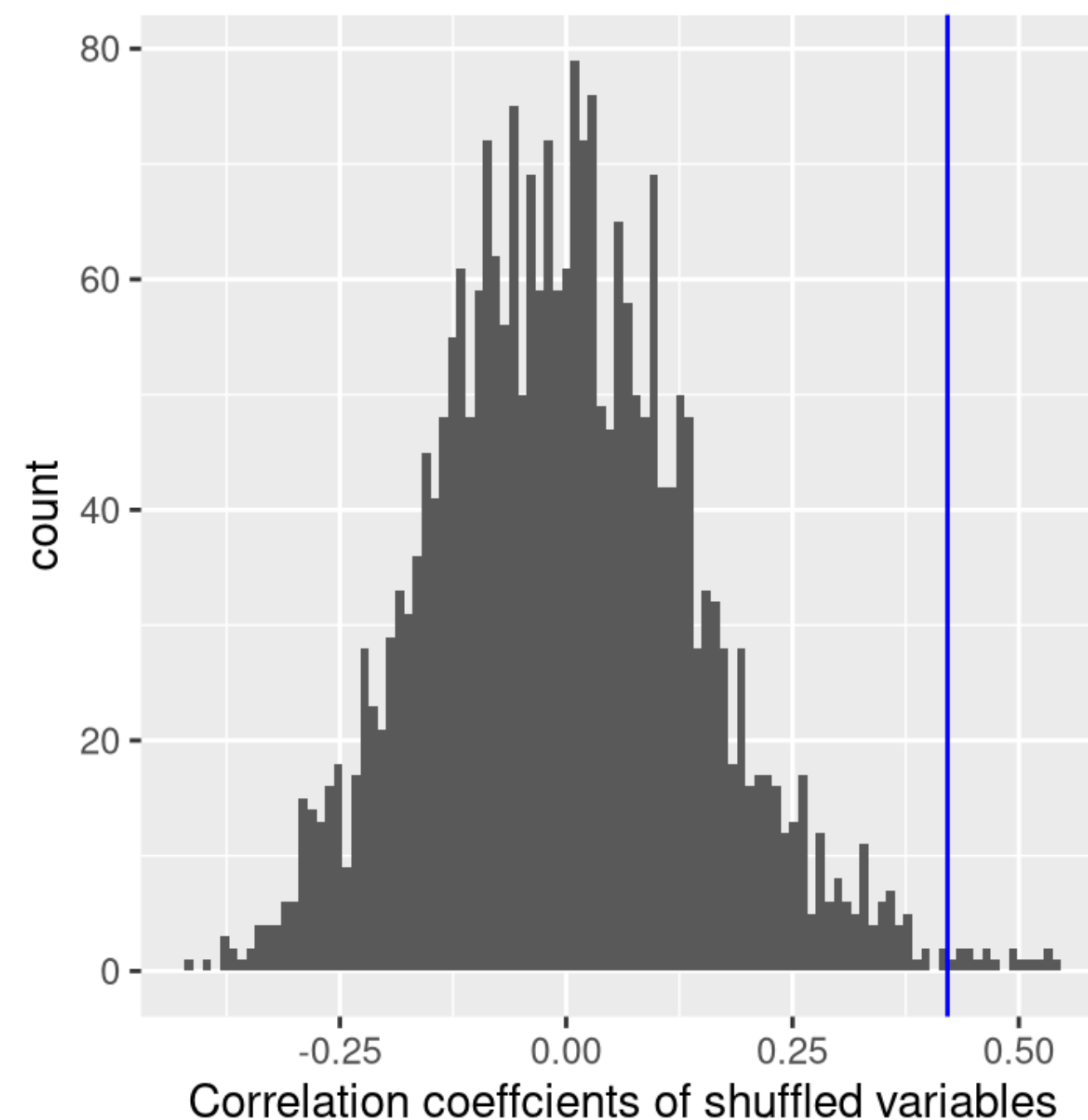
Hypothesis testing for correlations

We can test null hypothesis that correlation is zero, by easily converting correlation value into a t-statistic:

$$t_r = \frac{r\sqrt{N-2}}{\sqrt{1-r^2}}$$

We find that the five thirty data's correlation is 0.42, and p-value = 0.002

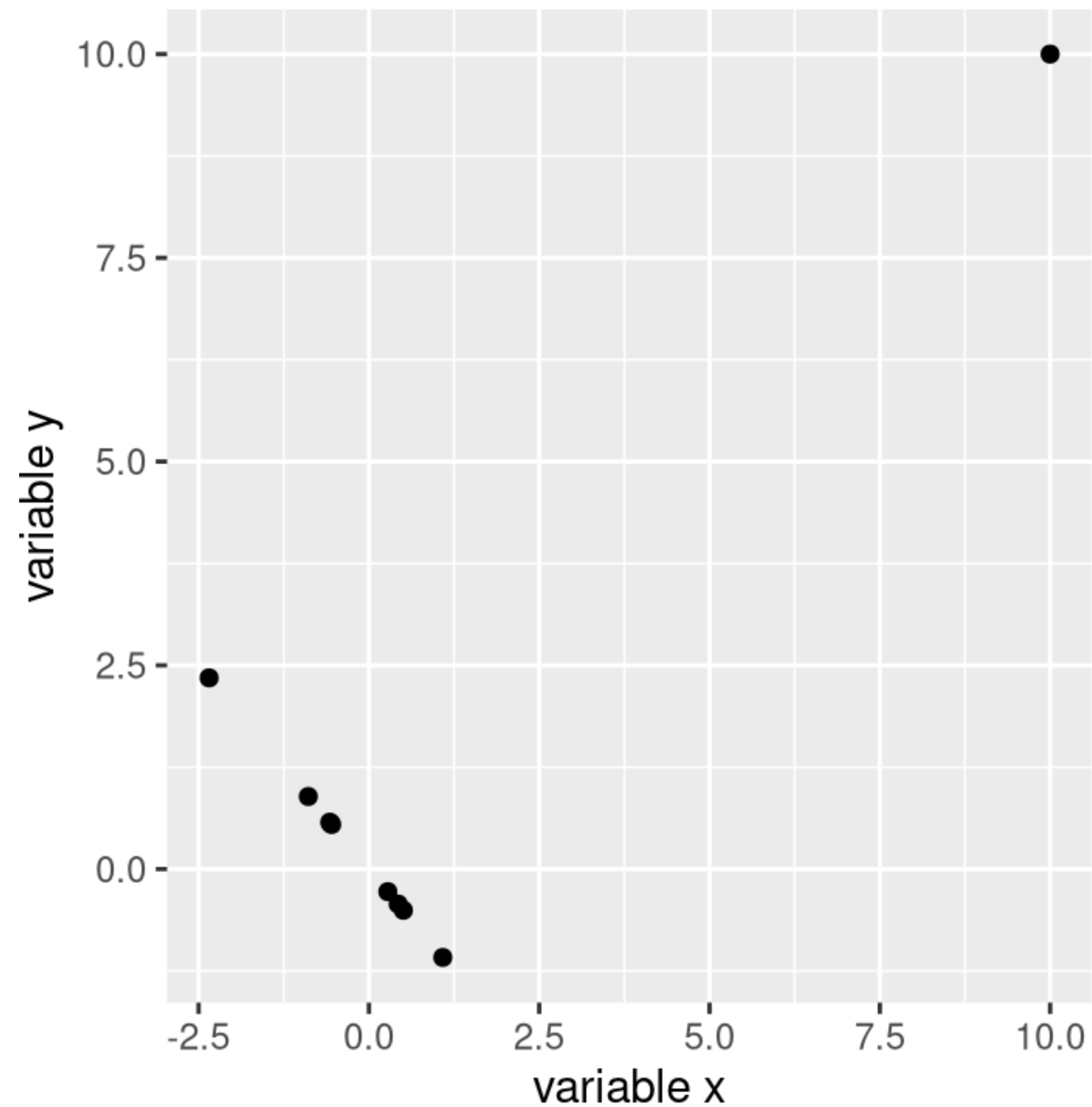
p(shuffled r >= observed) = 0.007



Note: we can also do this by sampling i.e., repeatedly shuffling the data and drawing a histogram of correlation values

Sidenote: outliers and correlations

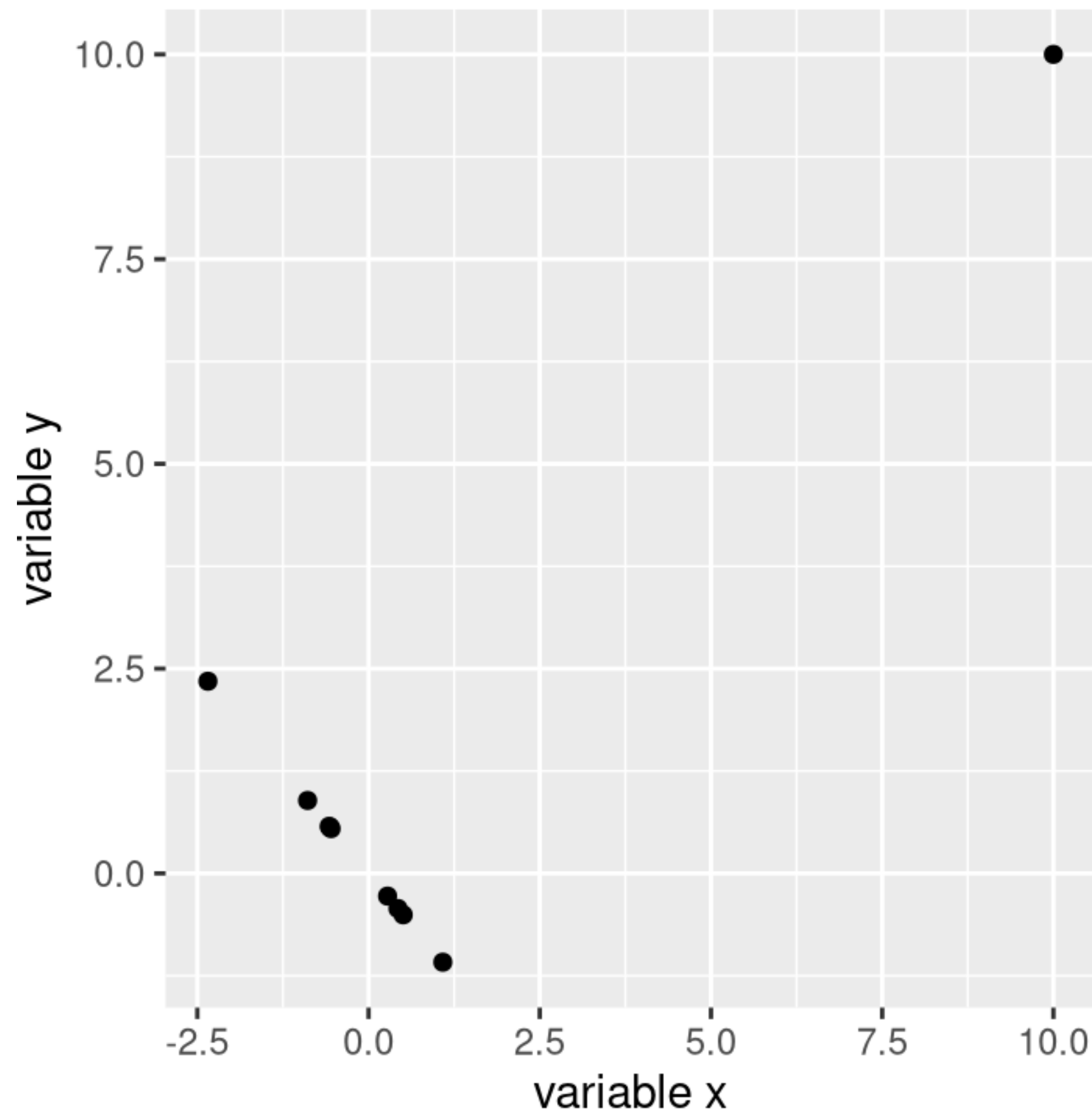
$r = 0.83$ (without outlier: $r = -1.00$)



How do we fix this?

Spearman's rank correlation

$r = 0.83$ (without outlier: $r = -1.00$)



How do we fix this?

Compute correlation on the ***ranks*** of the data after ordering them, rather than on the data themselves

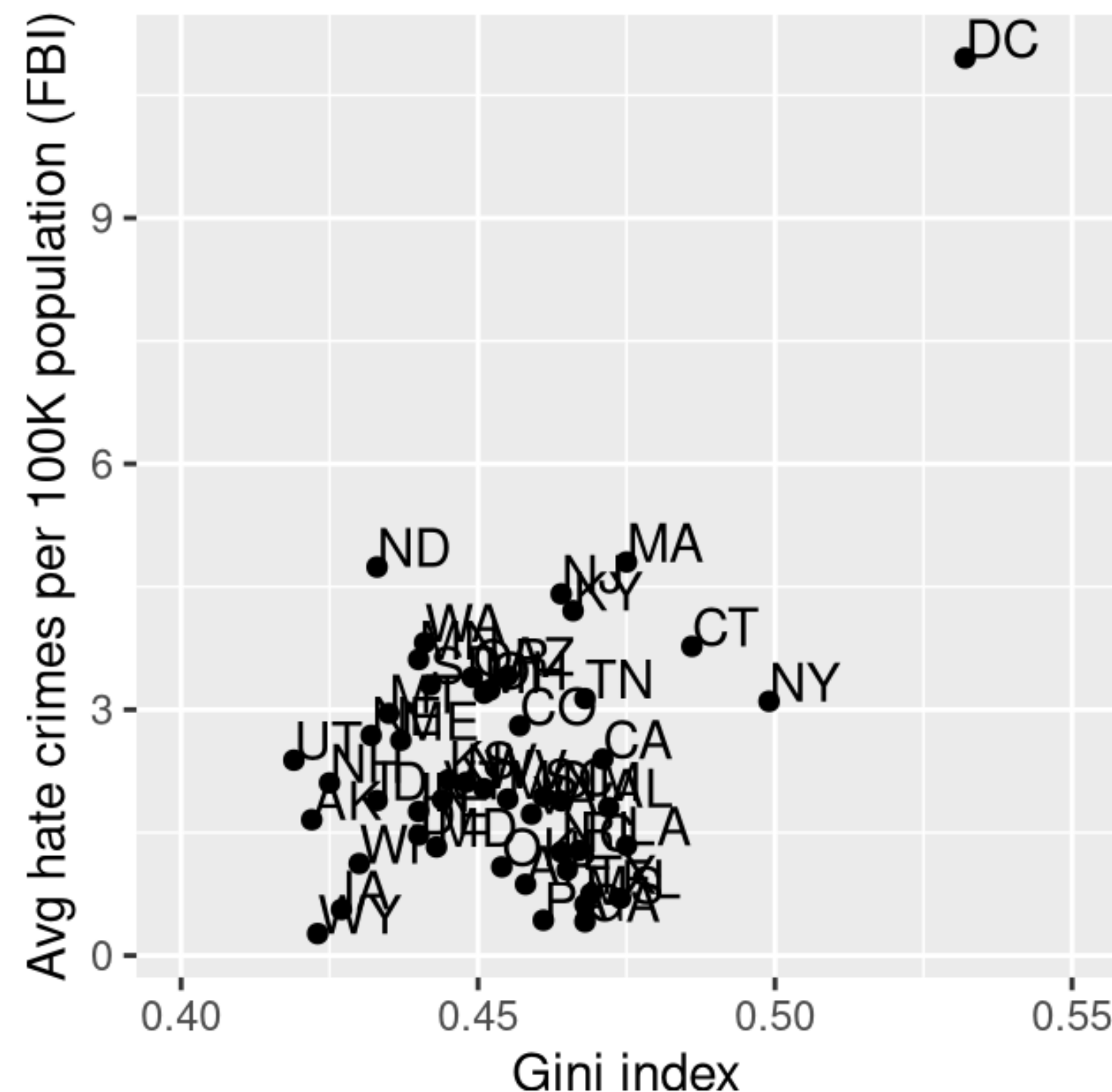
..so an outlier point of 1500 would be weighted same as 15

Spearman correlation = -0.45 on this data

Returning back to our example..

Higher Rates Of Hate Crimes Are Tied To Income Inequality (Fivethirtyeight.com, 2017)

“We found that income inequality was the most significant determinant of population-adjusted hate crimes and hate incidents across the United States”.



Is income inequality related to hate crimes?

When we use spearman, results are insignificant!

Spearman correlation = 0.033, p-value = 0.8

Pearson vs. spearman

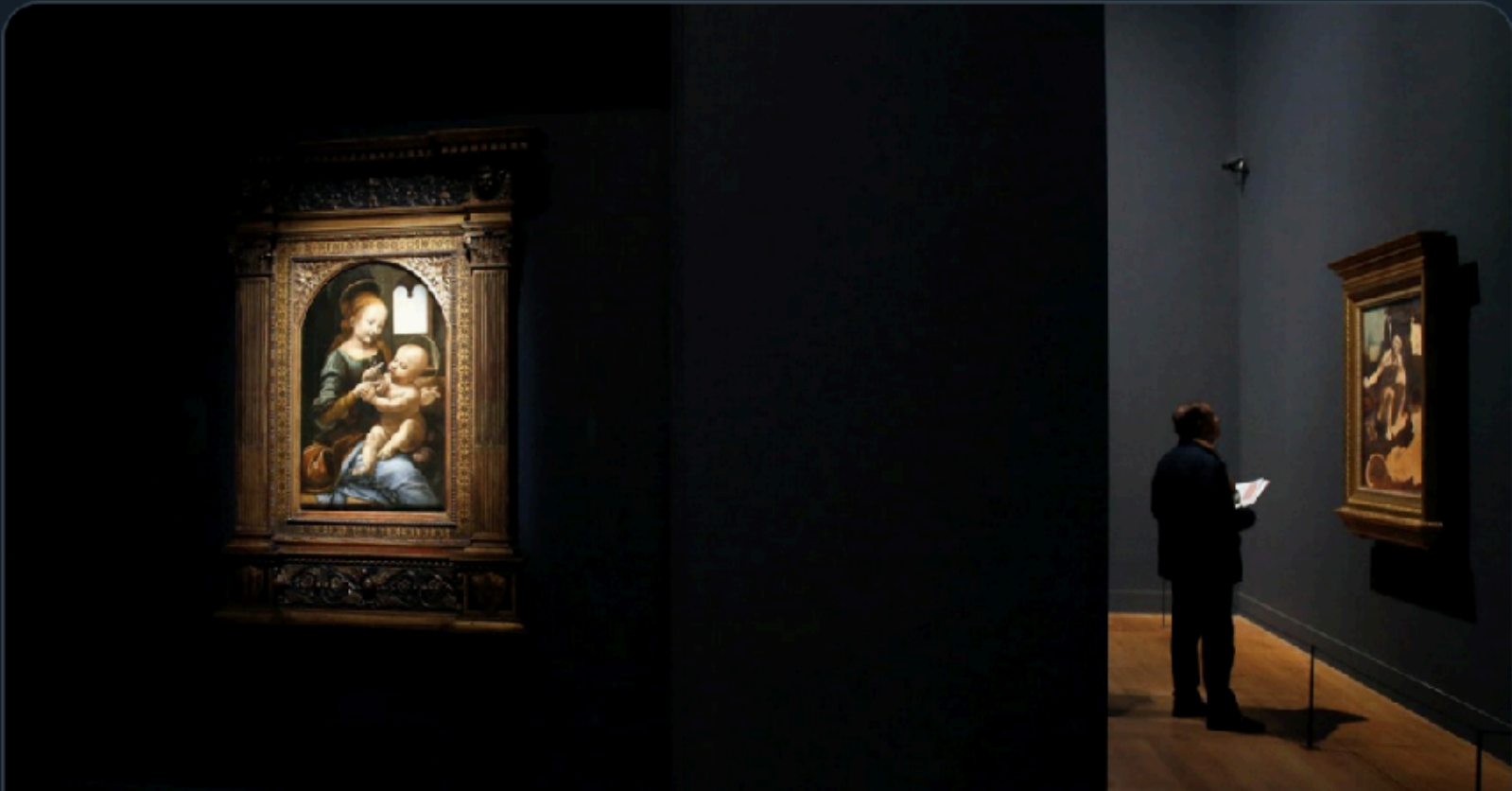
- What's better?
 - depends on the context
 - Spearman is robust to outliers, but it throws away (potentially useful) information

Correlation is NOT causation



NYT Health
@NYTHealth

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.



Another Benefit to Going to Museums? You May Live Longer

Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.

[nytimes.com](https://www.nytimes.com)

9:19 AM · Dec 22, 2019 · [SocialFlow](#)

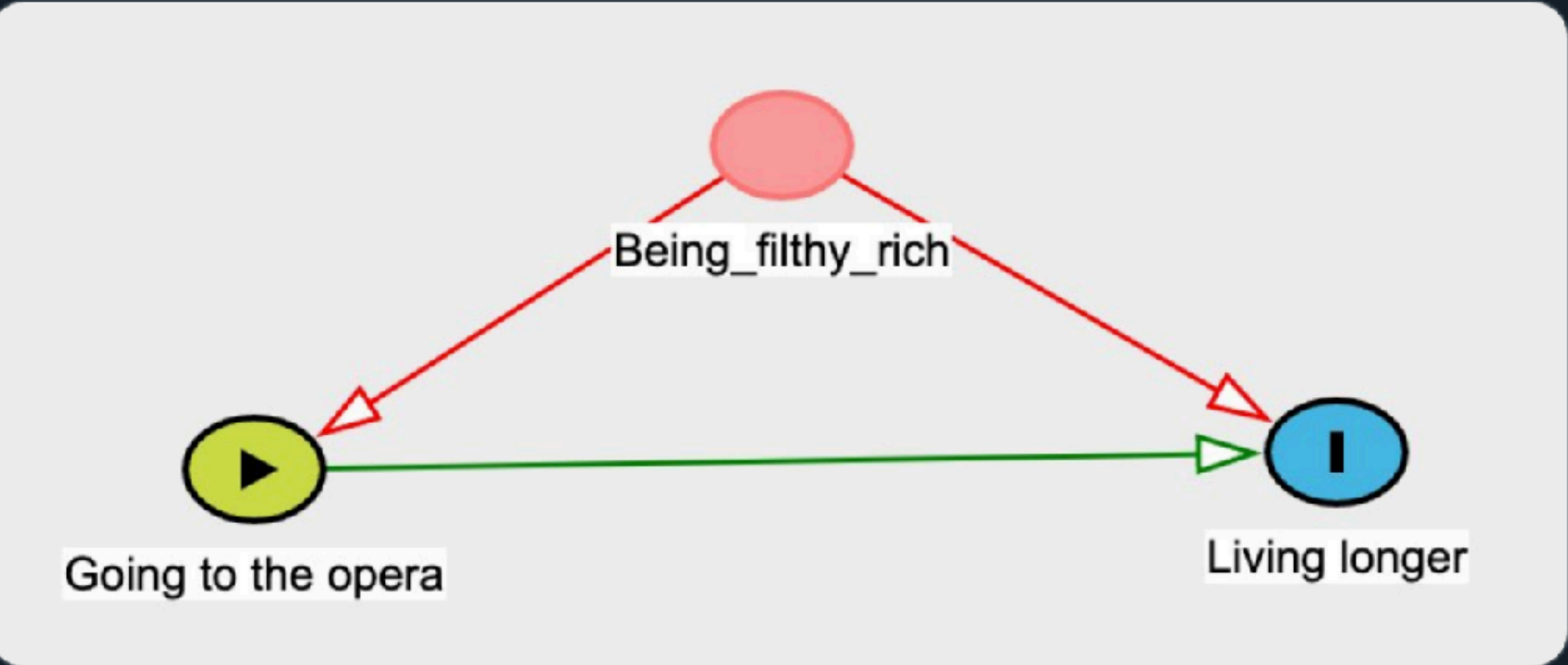
336 Retweets

1.3K Likes



Andrew Heiss
@andrewheiss

ooh ooh i can draw the dag for this one!



```
graph TD; A((Being filthy rich)) --> B((Going to the opera)); A --> C((Living longer)); B --> C;
```

 **NYT Health** @NYTHealth · Dec 22, 2019

Want to live longer? Try going to the opera. Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't. nyti.ms/2Q9AmZV

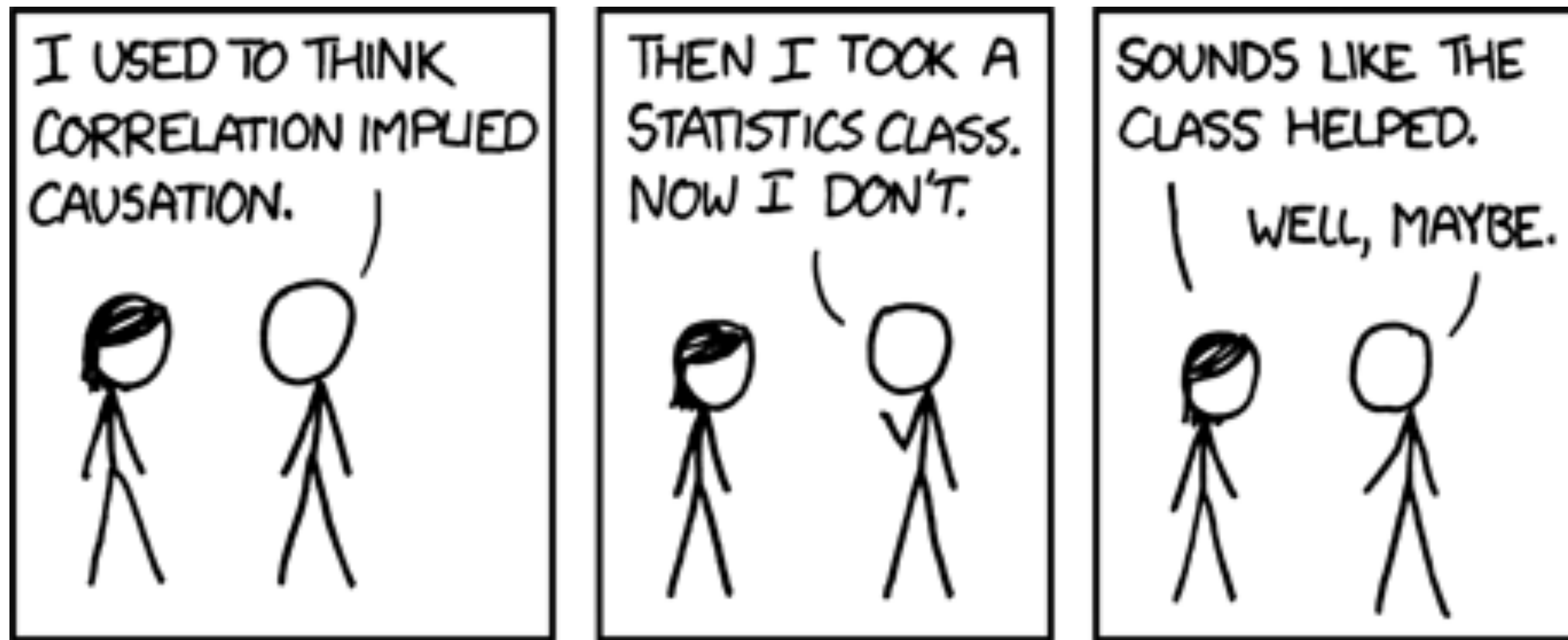
2:47 PM · Dec 22, 2019 · [Twitter Web App](#)

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- correlations suggest that there is *some* causal relationship (maybe)
- but this relationship need not be a direct causal relationship from A to B (or from B to A)